Day 2 Exercises: Problem Solving Sessions 1 and 2

This handout combines both Day 2 problem-solving sessions.

Session 1

This session develops fluency with vector operations, matrix-vector products, and dimension checks. Focus on correct setup before arithmetic.

1. Compute $\begin{bmatrix} 3 \\ -1 \end{bmatrix} + \begin{bmatrix} 4 \\ 2 \end{bmatrix}$.
2. Compute $2\begin{bmatrix} 1 \\ 5 \end{bmatrix} - \begin{bmatrix} 3 \\ 4 \end{bmatrix}$.
3. For $A = \begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix}$ and $\mathbf{x} = \begin{bmatrix} 4 \\ 2 \end{bmatrix}$, compute $A\mathbf{x}$.
4. Is AB defined if A is 2×3 and B is 3×2? What size is AB?
5. Is BA defined for the same matrices in Problem 4? What size?
6. Interpret entry a_{21} in words.
7. Give one real meaning of a 2-entry vector in school life.
8. Give one real meaning of a 2x2 matrix.
9. Let $M = \begin{bmatrix} 1 & 4 & 2 \\ 0 & 3 & 5 \end{bmatrix}$ and $\mathbf{u} = \begin{bmatrix} 2 \\ 1 \\ -1 \end{bmatrix}$. Compute $M\mathbf{u}$.
10. Explain why matrix multiplication order matters by giving dimensions where AB is defined but BA is not.

Session 2

This session practices building and solving systems from real contexts. Focus on interpreting each equation and checking solutions.

1. Solve: $\begin{cases} x + 2y = 9 \\ 3x - y = 5 \end{cases}$.

2. Write a 2-equation system for two snack types with known total count and total cost.

3. A matrix product is not defined. What dimension check should you do first?

4. Explain in one sentence how vectors and systems are connected.

5. Ticket model: total 200 tickets, prices 6 and 10, revenue 1520. Write (do not solve) a system.

6. Solve by substitution: $\begin{cases} 2x + y = 13 \\ x - y = 2 \end{cases}$.

7. Solve by elimination: $\begin{cases} 3x + 2y = 16 \\ 3x - 2y = 8 \end{cases}$.

8. Cafeteria model: sandwiches s and salads d. Total items: 90. Prices: \$5 and \$7. Total revenue: \$570. Write and solve the system.

9. Reflection: in one sentence, describe one assumption in Problem 8 and how it could fail in reality.